Business Problem

Retail Store has experienced inconsistent sales and declining profitability across different regions and product categories. Management cannot easily identify which products, customers, and regions are driving revenue and which are reducing profits. They need Analysis of sales, performance of last year, and support data-driven decisions to increase revenue and profitability.

The goal of this analysis is to:

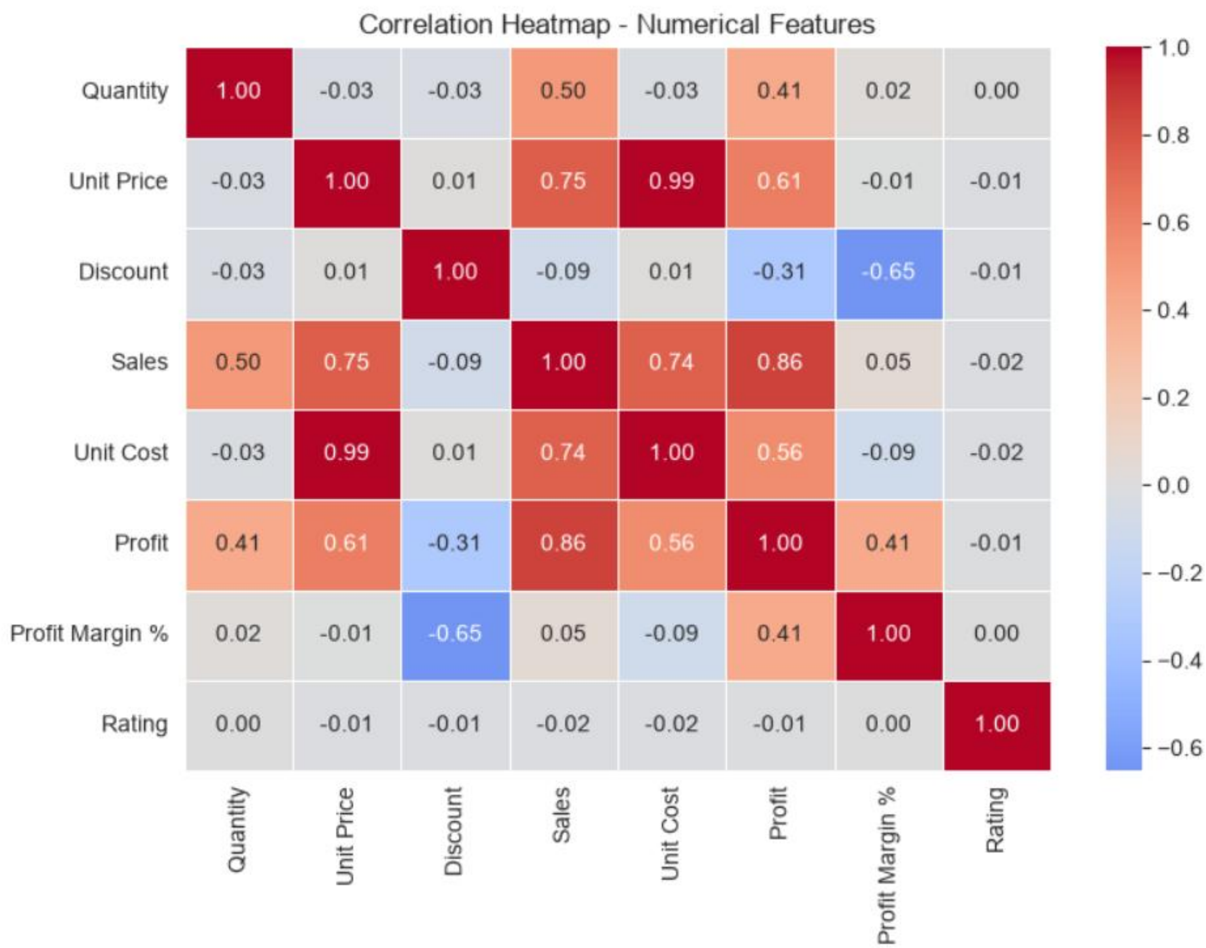
- Identify underperforming products that require promotional or pricing adjustments.
- Determine top customers contributing to sales and gross profit.
- Analyze top products by sales.
- Identify low-profit products
- Investigate the profitability variance between high-performing and low-performing products.

Exploratory Data Analysis Insights

Summary Statistics

	count	mean	min	25%	50%	75%	max	std
Order Date	2483	2024-03-27 00:40:00.966572	2023-01-01 00:00:00	2023-08-13 00:00:00	2024-04-04 00:00:00	2024-11-09 00:00:00	2025-06-19 00:00:00	NaN
Ship Date	2483	2024-03-30 23:29:15.779299	2023-01-04 00:00:00	2023-08-17 00:00:00	2024-04-07 00:00:00	2024-11-12 12:00:00	2025-06-24 00:00:00	NaN
Quantity	2483.0	4.5441	1.0	3.0	5.0	7.0	8.0	2.301238
Unit Price	2483.0	370.628087	31.5	175.305	259.0	549.3	919.58	280.787528
Discount	2483.0	0.102416	0.0	0.05	0.1	0.15	0.2	0.069578
Sales	2483.0	1492.997008	26.35	301.165	1003.11	2118.695	7302.48	1480.09187
Unit Cost	2483.0	263.514978	19.11	121.655	184.9	404.1	736.48	201.908857
Profit	2483.0	310.667672	-84.5	50.93	170.7	433.25	2775.04	373.183979
Profit Margin %	2483.0	20.327503	-2.49	13.96	20.56	27.555	39.97	9.449063
Year	2483.0	2023.779702	2023.0	2023.0	2024.0	2024.0	2025.0	0.729956
Week	2483.0	24.247684	1.0	11.5	23.0	37.0	52.0	15.046218
Rating	2483.0	4.001208	3.0	3.0	4.0	5.0	5.0	0.816578

Correlation Insights



1. Unit Price vs. Unit Cost: Strong correlation (0.99), indicating that product pricing closely follows the underlying unit cost.

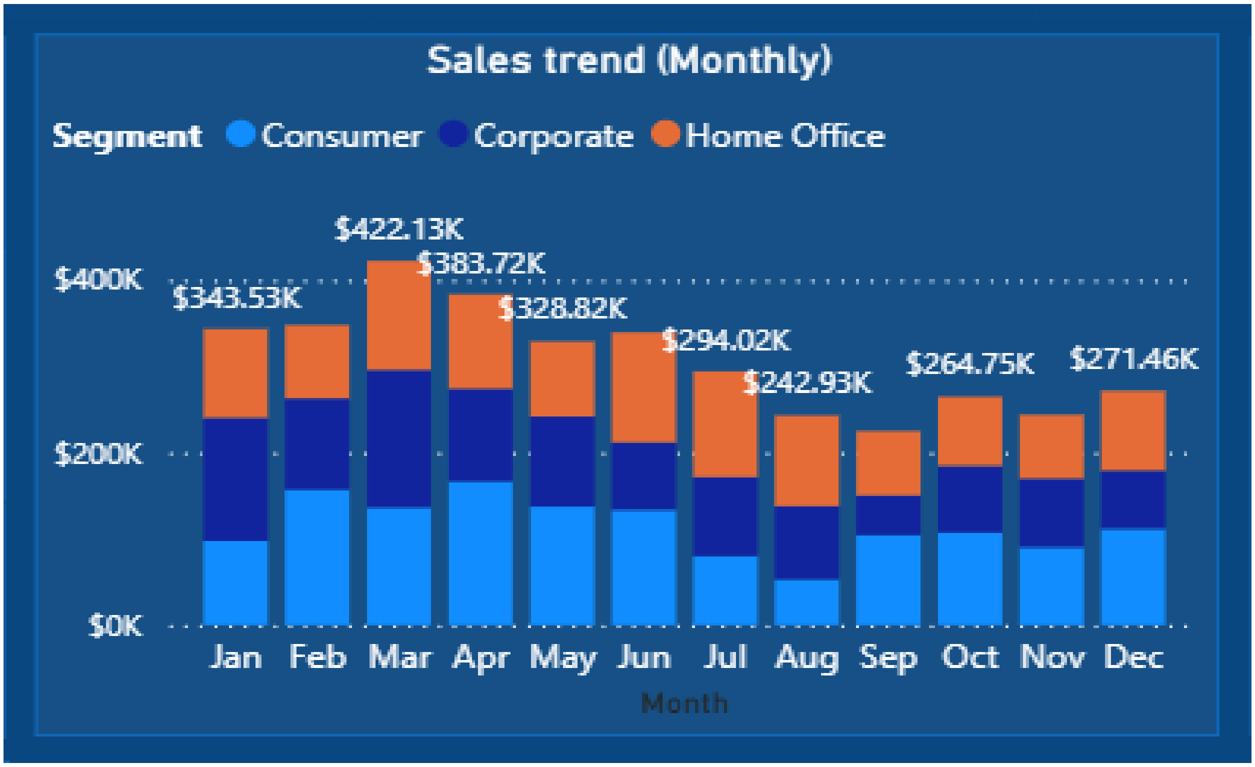
2. Sales vs. Profit: Strong correlation (0.86), confirming that higher sales are strongly associated with higher profitability.

3. Discount vs. Profit Margin: Negative correlation (-0.65), suggesting that increasing discounts may reduce profit margins.

4. Unit Price vs. Sales: Strong correlation (0.75), indicating that higher-priced products tend to generate higher sales revenue.

Research Questions & Key Findings

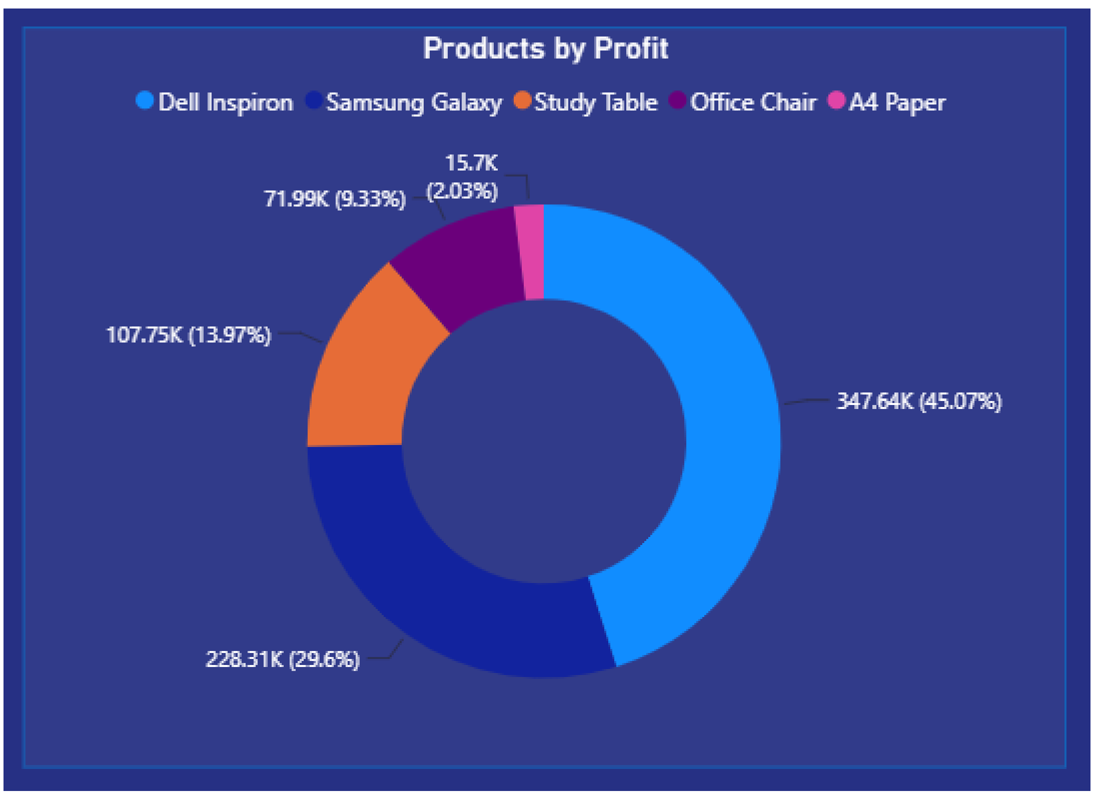
1. Monthly Sales Overview



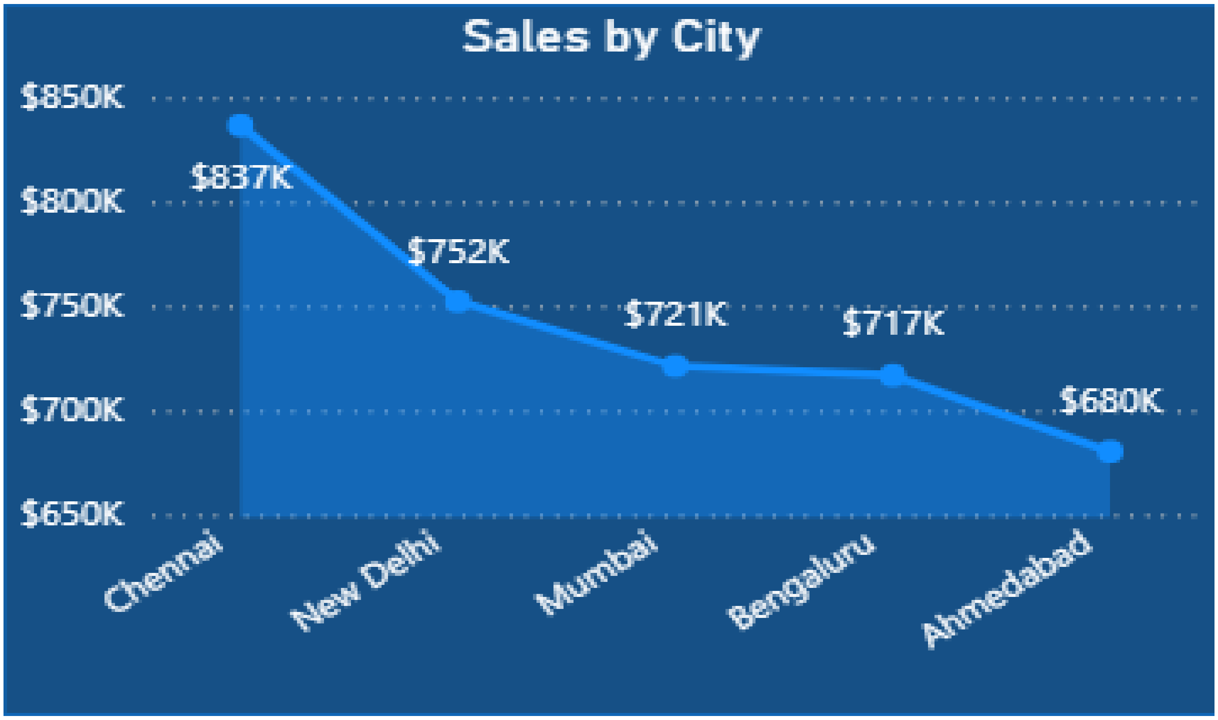
- The chart shows the monthly sales trend from January to December, divided into Consumer, Corporate, and Home Office segments.
- March has the highest sales at approximately **$422.13K.**
- Sales gradually decline from **April** to **August**, reaching around **$242.93K** in July.
- Sales begin recovering from September, reaching about **$271.46K** in December.
- Overall, the chart indicates strong sales in the first half of the year, followed by a decline and moderate recovery toward year-end.

2. Products Profit

- The donut chart shows profit generated by different products.
- Dell Inspiron contributes the highest profit at **347.64K** (45.07%).
- Samsung Galaxy is the second-highest contributor with **228.31K** (29.6%).
- Study Table and Office Chair contribute **107.75K** (13.97%) and **71.99K** (9.33%), respectively.
- A4 Paper has the lowest profit at **15.7K** (2.03%), indicating the smallest contribution.



3. Sales by City



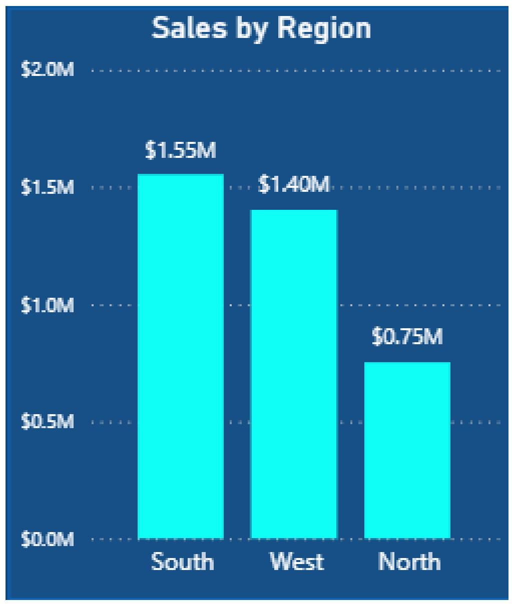
- The Sales by City chart compares sales performance across five major cities.
- Chennai has the highest sales at approximately **$837K**, followed by New Delhi at **$752K**.
- Mumbai and Bengaluru show similar sales levels of about **$721K** and **$717K**.
- Ahmedabad has the lowest sales at approximately **$680K** among the five cities.
- Overall, Chennai is the strongest-performing city, while Ahmedabad has the most scope for improvement.

4. Return Rate% by Category

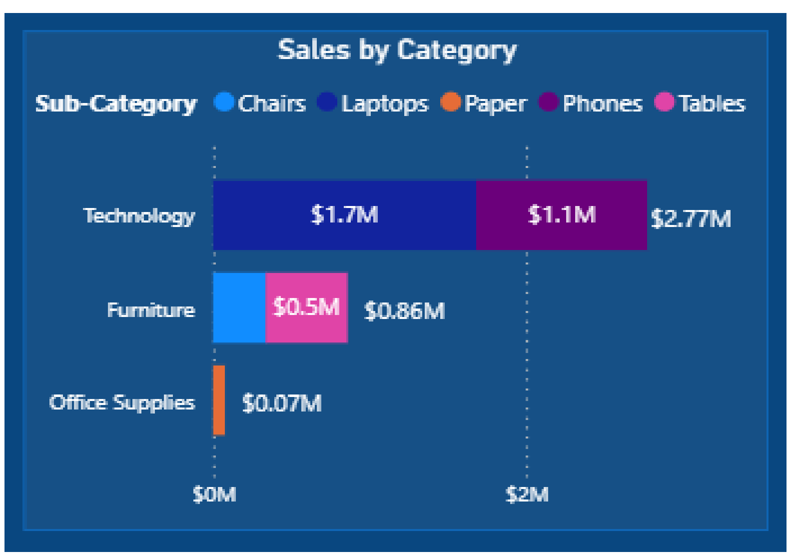
- The chart shows the return rate of orders across three product categories: Furniture, Office Supplies, and Technology.
- Furniture has approximately 52% non-returned and 48% returned orders.
- Office Supplies has about 49% non-returned and 51% returned orders.
- Technology has around 52% non-returned and 48% returned orders.
- Overall, Office Supplies has the highest return rate, indicating a potential area for improvement.

5. Total Sales by Region

- The Sales by Region chart compares sales performance across three regions: South, West, and North.
- The South region has the highest sales at approximately $1.55M.
- The West region follows with sales of around $1.40M.
- The North region has the lowest sales at approximately $0.75M.
- Overall, the South and West regions perform strongly, while the North region has significant potential for growth.



6. Sales by Category



- The Sales by Category chart compares sales across Technology, Furniture, and Office Supplies.
- Technology generates the highest sales at approximately $2.77M, mainly from Laptops and Phones.
- Furniture records around $0.86M, with Tables and Chairs contributing significantly.
- Office Supplies has the lowest sales at approximately $0.07M.

Final Recommendations

1. Protect margin: reduce unnecessary discounting, especially where markdowns do not generate enough incremental sales.

2. Scale top products: support high-performing products with focused marketing and sufficient inventory.

3. Strengthen weaker categories: investigate why Office Supplies trails the other categories and test targeted offers.

4. Prioritize strong regions: maintain inventory and marketing support in South and other high-performing regions.

5. Validate returns: investigate the approximately 50% return rate before using it as a customer-behavior KPI.

6. Prepare for seasonality: use monthly sales patterns to plan inventory and staffing ahead of stronger demand periods.